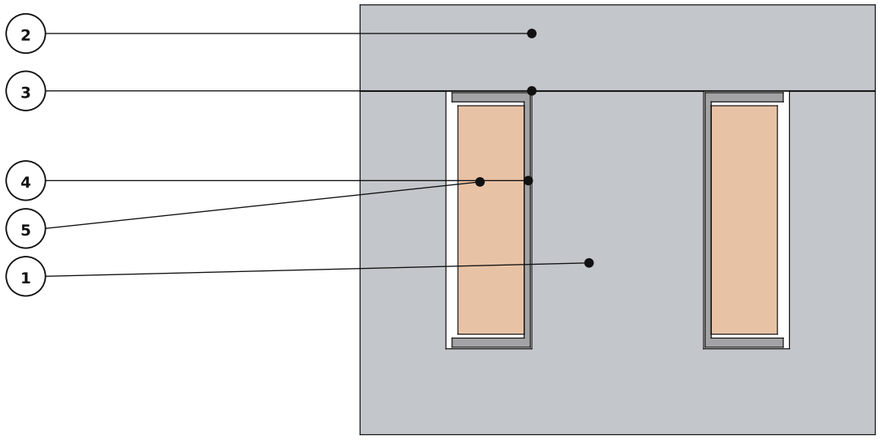
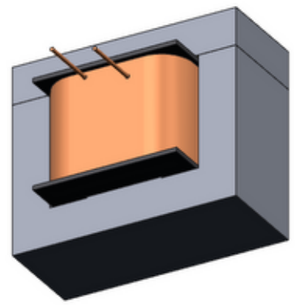
SECTION A–A: the cut through the stack's middle (seen from +z, the legs up) 1:1



REVISIONS		
Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

PARTS LIST

item	drawing	description	material	qty
1	P-019	E laminations, a bonded stack	M400-50A 0.5	1
2	P-019	I laminations, a bonded stack	M400-50A 0.5	1
3	P-019	gap spacer, 0.063 mm	PET film	1
4	P-019	bobbin	PA66-GF	1
5	REL-721	winding, 150 turns	Cu, grade 2	1
6	REL-721	leads, sleeved	Cu	2



3-D view, a quarter cut away; the reference parts left out (axonometric, not to scale)

NOTES

1. La and Lb are one part, two per machine: EI-84 scrapless (84 × 70, centre leg 28, outer legs 14, window 14 × 42), a 35 mm stack of M400-50A 0.5 mm laminations (M235-35A roughly halves the iron loss).
2. 150 turns of Ø1.40 grade 2 in 7 layers of 24: R20 0.276 Ω; 0.146 H, trimmed on test by the spacer (0.063 mm in all three legs, E and I butted).
3. Bond the stack and the spacer: the gap pulls 531 N at the peak, 226 N pulsating at 120 Hz. Impregnate (VPI, class F).
4. On the rotor at r 80 it sees 32 g (490 N at 600 rpm; 760 N in the 750 rpm spin test): La and Lb 180° apart in one plane; no bracket may encircle a leg (a shorted turn). Its place on the rotor is OPEN.
5. 1.66 kg each: 1.20 kg of iron, 0.34 kg of copper, the bobbin and the varnish. The flanges reach 1 mm past the build: 84 × 70 × 61 over all.

DCCREG turbine · the magnetic pump: La / Lb

La / Lb choke, EI-84 × 35, 150 turns		Drawing no.
Material:		**DCCREG-REL-720**
Mass: —	Qty per machine: 2	Rev A Sheet 59 of 103 A3
Scale: 1:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**